
Agentic Job Market Intelligence: A Dynamic Graph Approach for Student Career Guidance and Skill Gap Analysis

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Declaration

We do hereby declare that the research works presented in this thesis entitled, “Thesis/Report Title” are the results of our works. We further declare that the thesis/report has been compiled and written by us, and no part of this thesis/report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.

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Approval

I do hereby declare that the research works presented in this thesis entitled, “**Agentic Job Market Intelligence: A Dynamic Graph Approach for Student Career Guidance and Skill Gap Analysis**”, are the outcome of the original works carried out by the **Hasibul Hasan Shanto, Nazia Jafrin, Rubaiya Islam Rimly, Humayra Tasnim**, under my supervision. I further declare that no part of this thesis/report has been submitted elsewhere for the requirements of any degree, award, diploma, or other purposes except for publications. I further certify that the dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

We would like to dedicate this research to our loving parents . . .

Acknowledgment

We want to express our heartfelt gratitude to the almighty Allah who offered our family and us kind care throughout this journey. We are deeply thankful to the Bangladesh University of Business and Technology (BUBT) for providing us with such a wonderful environment to pursue our research/project. We would like to express our sincere gratitude to **Md. Shahiduzzaman**, Assistant Professor, Department of CSE, BUBT. We have completed our research with his help. We found the research area, topic, and problem with his suggestions. He guided us with our study and supplied us with many research papers and academic resources in this area. He is patient and responsible. When we had questions and needed his help, he would always find time to meet and discuss with us no matter how busy he was. We also want to give thanks to **Md. Saddam Hossain**, Assistant Professor, Department of CSE, and all our teachers for providing a solid background for our studies and research thereafter. They have been a great source of inspiration to us, and we thank them from the bottom of our hearts. We would also like to acknowledge our team members for supporting each other and be grateful to our university for providing this opportunity for us.

Abstract

The increasing gap between academic education and the dynamic demands of the global job market has become a major challenge for university students, often resulting in underemployment and delayed career entry. Traditional career guidance systems fail to provide personalized, real-time, and actionable insights aligned with current industry trends. To address this issue, this project proposes Agentic Job Market Intelligence, a dynamic and intelligent platform designed to support student career development through advanced data-driven techniques. Agentic Job Market Intelligence leverages real-time job market data and represents it using a dynamic knowledge graph that models relationships between skills, job roles, and learning resources. The system integrates a multi-agent intelligence engine capable of performing complex reasoning, enabling personalized skill gap analysis, career path recommendations, and structured learning roadmaps. By combining graph-based modeling with AI-driven analysis, the platform ensures transparency, adaptability, and accuracy in its recommendations. The proposed system not only identifies the current skill level of users but also predicts future skill demands, helping students proactively prepare for evolving job markets. Additionally, Agentic Job Market Intelligence enhances user understanding by providing explainable recommendations, making the decision-making process more reliable and user-friendly. Overall, the system aims to bridge the gap between education and employment by offering a scalable, intelligent, and practical solution for career guidance and workforce readiness.

Keywords: Agentic, leverages, adaptability, proactive

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List of Abbreviations and Acronyms

HTML	Hyper Text Markup Language
CNN	Convolutional Neural Network
AI	Artificial Intelligence
KG	Knowledge Graph
LLM	Large Language Model
API	Application Programming Interface
UI	User Interface

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Chapter 1

1. Introduction

Globally, university students struggle to close the gap between their education and the changing demands of the job market, which frequently leads to underemployment or extended unemployment after graduation. Agentic Job Market Intelligence, an intelligent platform that uses real-time job market data structured as a dynamic knowledge graph, emerges as a creative solution to this problem. Agentic Job Market Intelligence provides individualized skill gap analyses, sequenced learning roadmaps, and customized career recommendations by combining graph-based reasoning with a multi-agent intelligence engine. These services are based on verified market trends and are created for maximum transparency and actionability.

1.1. Problem Statement

The fundamental problem is the mismatch between students' self-reported abilities and the specific demands of employers, which is made worse by out-of-date or generic career counseling programs that don't take into account current job market data. Students often enter the workforce unprepared, with no obvious path to remediation or prioritizing, and are repeatedly rejected because of undiagnosed skill deficits. In order to address this, Agentic Job Market Intelligence combines real-world job data into an understandable framework that allows for accurate diagnostics and strategic planning that are not possible with traditional methods.

1.2. Problem Background

The fundamental problem is the mismatch between students' self-reported abilities and the specific demands of employers, which is made worse by out-of-date or generic career counseling programs that don't take into account current job market data. Students often enter the workforce unprepared, with no obvious path to remediation or prioritizing, and are repeatedly rejected because of undiagnosed skill deficits. In order to address this, Agentic Job Market Intelligence combines real-world job data into an understandable framework that allows for accurate diagnostics and strategic planning that are not possible with traditional methods.

1.3. Research Objectives

- I. An agentic AI system that autonomously collects and refreshes job market skill data without manual intervention.
- II. A dynamic knowledge graph that continuously updates job–skill relationships and skill demand weights over time.
- III. Graph-based reasoning that explains skill importance through structured relationships instead of keyword similarity.
- IV. A quantitative “Distance to Employment” metric that measures how far a student is from market readiness.
- V. Real-time detection of emerging skills and technologies using continuous job market signals.

1.4. Motivations

The urgent need to empower students in settings with limited resources, like those at BUBT, where access to high-quality career tools is restricted, is what motivates this effort. The drive to develop an open, deployable system that democratizes high-quality career knowledge is fueled by firsthand observations of colleagues having difficulty finding employment as well as holes in the openness and personalization of current platforms.

1.5. Significance of the Research

By decreasing skill mismatches, increasing graduate employability, and incorporating empirical market data into educational courses, CareerGraph has the potential to be a game-changer. By matching talent with labor demands in fast-paced industries like technology and data, its emphasis on explainability builds student trust and its scalable architecture facilitates deployment across universities, ultimately boosting economic efficiency.

1.6. Research Contribution

Significant contributions include a customized 10,000-posting dataset for repeatable validation, a modular multi-agent engine that combines algorithmic precision with LLM natural language creation, and a novel Neo4j-based knowledge graph schema for skill-job links. These developments provide a model for transparent, graph-powered recommendation systems that may be tailored to a variety of industries, advancing agentic AI applications in education.

1.7. Complex Engineering Analysis

1.7.1. Complex Engineering Problems

- I. WP1 (Depth of knowledge required): Demands expertise in AI agents, graph databases, and job market semantics.
- II. WP2 (Range of conflicting requirements): Balances real-time updates, ethical data sourcing, and student privacy.
- III. WP3 (Depth of analysis required): Involves graph traversal for multi-hop reasoning (e.g., indirect skills like Python → JavaScript).
- IV. WP4 (Familiarity of issues): Builds on known skill gaps but innovates with agentic dynamism.
- V. WP5 (Extent of applicable codes): Complies with data privacy (GDPR-like) and open-source licensing.
- VI. WP6 (Extent of stakeholder involvement & conflicting requirements): Engages students, universities, and employers with varying priorities.
- VII. WP7 (Interdependence): Modules (collector, graph, analyzer) are tightly coupled, with failures propagating.

1.7.2. Complex Engineering Activities

- I. WK1 (Natural Sciences): Not directly applicable.
- II. WK2 (Mathematics): Applied in graph algorithms for skill similarity (e.g., shortest path for learning pathways) and demand weighting.
- III. WK3 (Engineering Fundamentals): Core for system architecture, data pipelines, and agent orchestration.
- IV. WK4 (Specialist Knowledge): Essential for AI agents (LangChain), knowledge graphs (Neo4j), and NLP skill extraction.
- V. WK5 (Engineering Design): Critical for dynamic graph schemas and adaptive recommendation logic.
- VI. WK6 (Engineering Practice): Required for ethical data handling, agent autonomy, and scalable deployment.
- VII. WK7 (Comprehension): Key for interpreting job market ethics and stakeholder impacts on student employability.
- VIII. WK8 (Research Literature): Used to benchmark against O*NET/ESCO and agentic AI papers.

Project Organization

This report is organized into three chapters, each addressing a specific aspect of the proposed system.

Chapter 1 provides an introduction to the project, including its objectives, scope, and motivation. It also presents an overview of the report structure.

Chapter 2 discusses the foundational aspects of the study. It includes the background study, which explains the fundamental concepts and relevant technologies. The literature review section analyzes existing works and related systems. The problem analysis section identifies the limitations of current approaches and defines the problem addressed in this project.

Chapter 3 presents the methodology of the proposed system. It describes the system architecture and provides a feasibility study, including economic, technical, and operational feasibility analyses. The requirement analysis section outlines both functional and non-functional requirements, along with the tools and technologies used. Additionally, the system design section includes various diagrams such as use case diagrams, context diagrams, data flow diagrams, entity-relationship diagrams, database schema diagrams, and system flowcharts to illustrate the overall design and workflow of the system.

Chapter 2

2. Background Study

The swift transformation of the global job market has resulted in a notable disconnect between academic training and the needs of industries. Conventional education systems typically adhere to fixed curricula that do not swiftly adjust to new technologies and evolving skill requirements. Consequently, numerous graduates enter the workforce lacking the essential skills sought by employers, which can lead to underemployment or postponed career starts. This challenge is especially pressing in rapidly expanding sectors like information technology, where new tools, frameworks, and job positions are constantly being introduced. Recent developments in artificial intelligence, machine learning, and data analytics have facilitated the creation of advanced career guidance systems. These systems are designed to evaluate user profiles, identify pertinent skills, and suggest appropriate career paths or educational resources.

Moreover, knowledge graphs have attracted interest for their capacity to illustrate intricate relationships among entities such as skills, jobs, and courses in a clear and organized way. By mapping these connections, knowledge graphs offer a robust basis for comprehending skill interdependencies and career shifts. Additionally, the rise of extensive job market data from online platforms has enabled the analysis of real-time skill demand trends.

However, many current systems either depend on outdated datasets or lack the ability to effectively merge various data sources. This shortcoming diminishes their capacity to deliver precise and timely recommendations. Thus, there is an increasing demand for a system that integrates real-time labor market insights, graph-based modeling, and intelligent reasoning to assist students in making well-informed career choices.

2.1. Literature Review

The study in [1] introduced a knowledge graph-based model that integrates university curricula, job vacancies, and MOOCs to bridge the gap between academic education and industry requirements. The system utilizes graph databases and embedding techniques to establish relationships between skills, courses, and job roles, enabling efficient skill matching, gap identification, and personalized course recommendations.

The study in [2] presents *Skill-Pulse AI*, a unified workforce intelligence platform combining demand forecasting, anomaly detection, and career optimization. It integrates multiple analytical techniques to monitor labor market trends and generate career insights, although its dependency on simulated datasets limits real-world validation.

The system in [3] developed a machine learning-based career guidance model using network visualization to map student skills to job roles. It analyzes resumes and user inputs to position individuals within a job network, improving decision-making through visual representation and enhancing career awareness.

The study in [4] proposed a knowledge graph-based recommender system that extracts data from multiple educational platforms to provide reskilling and upskilling recommendations. The system connects skills, job roles, and learning resources to generate structured and meaningful career guidance.

The study in [5] explored temporal knowledge graph embeddings to forecast future skill demand by modeling job-skill relationships over time. This approach enables the prediction of emerging skills and provides a better understanding of dynamic labor market trends.

The system in [6] introduced an AI-based recommendation framework that aligns educational resources with labor market requirements using topic modeling and filtering techniques. It enhances learning outcomes by suggesting relevant and high-quality resources tailored to user needs.

The study in [7] proposed a generative AI-driven career guidance system combining natural language processing and large language models for personalized recommendations. The system can infer implicit skills and generate adaptive career paths based on user profiles.

The study in [8] introduced an explainable AI framework for classifying job roles and analyzing skill requirements using advanced neural network techniques. It provides interpretable outputs, helping users understand the reasoning behind recommendations.

The system in [9] developed an AI-powered career advisor integrating collaborative filtering, content-based filtering, and deep learning methods. It aims to provide personalized recommendations, although it lacks comprehensive performance evaluation.

The study in [10] proposed a conceptual framework for identifying green skill gaps using labor market intelligence and standardized taxonomies. It highlights the importance of sustainability-focused skills but lacks implementation details.

The study in [11] introduced a multi-agent and large language model-based system for adaptive and sequential career planning. The system allows multiple intelligent agents to collaborate in generating personalized and context-aware recommendations.

The system in [12] developed an intelligent job recommendation model using machine learning to match candidate skills with job requirements. While effective in personalization, it faces challenges in incorporating real-time job market updates.

The study in [13] proposed a hybrid recommendation framework combining collaborative filtering and knowledge-based approaches to improve accuracy. By leveraging both user behavior and structured data, the system enhances recommendation quality.

The study in [14] introduced a deep learning-based job matching system capable of identifying complex relationships between skills and job roles. It improves prediction performance but requires large datasets for training.

The system in [15] developed a big data-driven platform using distributed technologies to analyze large-scale job market trends and skill demand. It efficiently processes large volumes of data to extract meaningful insights.

The study in [16] proposed a semantic skill extraction framework using natural language processing and ontology-based techniques. It improves the accuracy of extracting relevant skills from resumes and job descriptions.

The study in [17] introduced a personalized learning recommendation system based on reinforcement learning to adapt learning paths dynamically. The system continuously updates recommendations based on user progress and feedback.

The system in [18] developed a graph-based career path recommendation model linking skills, jobs, and learning resources. It improves explainability and helps users understand possible career transitions.

The study in [19] proposed an AI-driven employability prediction system to evaluate student readiness and identify skill deficiencies. It helps institutions and learners take proactive steps to improve employability.

The study in [20] introduced a cloud-based career guidance platform that integrates real-time job market data and analytics for scalable recommendations. However, maintaining data freshness and consistency remains a key challenge.

It is clear from the studies that notable advancements have been achieved in career recommendation systems, skill gap analysis, and labor market intelligence. Nevertheless, many current systems do not offer a cohesive strategy that integrates real-time job market information, dynamic knowledge graph modeling, and multi-agent intelligence. This gap drives the creation of Agentic Job Market Intelligence, which seeks to deliver clear, adaptable, and tailored career guidance through a dynamic graph-based framework.

2.2. Problem Analysis

Numerous career recommendation systems have emerged, yet several key challenges persist. A major concern is the absence of real-time adaptability. Many of these systems rely on static or pre-existing datasets, which fail to capture the ever-evolving demands of the job market. Consequently, the suggestions made by these systems can quickly become outdated and less applicable.

Another critical issue is the insufficient personalization in career guidance. While some systems gather basic user information or historical data, they often do not encompass the full profile of a student, including implicit skills, learning styles, and career aspirations. This results in broad recommendations that may not adequately meet individual needs or address skill deficiencies. Furthermore, many current methodologies do not fully exploit the advantages of graph-based modeling.

Although some systems utilize knowledge graphs, they tend to be static and lack the ability to update dynamically. This limitation hinders their capacity to illustrate the changing relationships among skills, job roles, and educational resources. Additionally, the lack of multi-agent intelligence restricts the system's capability to engage in complex reasoning and offer detailed career planning.

Another significant hurdle is the deficiency in transparency and explainability of recommendations. Many AI-driven systems function as black boxes, making it challenging for users to comprehend the rationale behind suggested career paths or skills.

This diminishes user trust and hampers practical implementation. Thus, there is a pressing need for a more sophisticated solution that can tackle these challenges. A system like Agentic Job Market Intelligence is designed to address these issues by incorporating real-time job market data, dynamic knowledge graph modeling, and multi-agent intelligence to deliver personalized, transparent, and actionable career guidance, along with effective skill gap analysis.

Chapter 3

3. Methodology

Our project is an agent-based career intelligence platform designed to bridge the gap between university students' current skill profiles and the real demands of the modern job market. The methodology underpinning this system combines knowledge graph reasoning, algorithmic analysis, and optional large language model (LLM) integration to deliver personalized, explainable, and actionable career guidance.

The development process follows a **Test-Driven Development (TDD)** approach embedded within an iterative Agile workflow. A full backend with 87 passing tests was built before any frontend work began, ensuring a stable and well-defined API contract. The system design is informed by an extensive literature review of over 20 papers covering career recommendation systems, knowledge graphs for skill modeling, explainable AI, and labor market intelligence.

Rather than relying on opaque machine learning models, CareerGraph employs a transparent, multi-agent architecture. Four pure algorithmic agents — operating without any LLM dependency — compute skill gap scores, ranked job recommendations, learning roadmaps, and market demand trends using well-understood graph algorithms (Jaccard similarity, BFS, Kahn's topological sort). An optional LLM-powered Reasoning Agent sits on top of these results to generate natural language explanations, degrading gracefully to template-based responses when no LLM provider is configured.

This chapter details the system architecture, feasibility analysis, requirements specification, and complete design artifacts for the AI Driven platform.

3.1. System Architecture

CareerGraph is structured as a three-tier architecture: a **Presentation Layer** (React frontend), an **Application Layer** (FastAPI backend with a multi-agent Intelligence Engine), and a **Data Layer** (dual-database: PostgreSQL + Neo4j).

Presentation Layer — A React 18 single-page application built with TypeScript and Vite. It communicates exclusively via REST API calls, using TanStack Query for server-state caching and Axios for HTTP transport.

Application Layer — A FastAPI application exposing 13 REST endpoints across 8 resource groups, all prefixed under `/api/v1`. The Intelligence Engine is composed of seven specialized agents, each with a single responsibility, coordinated by the EngineOrchestrator.

Data Layer — A dual-database strategy: PostgreSQL stores structured relational data (users, authentication, profiles) with ACID guarantees; Neo4j stores the semantic knowledge graph (skills, jobs, prerequisites, courses) enabling graph traversal queries that are expensive or impossible in a relational model.

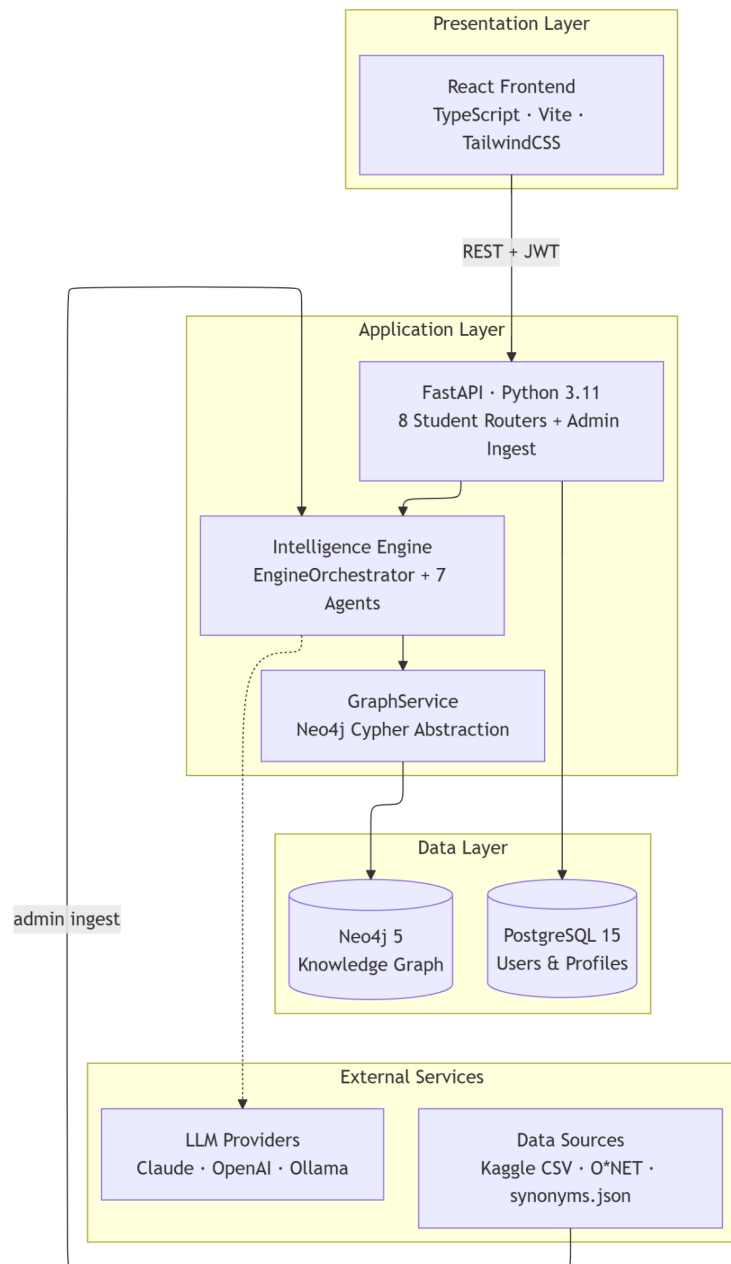


Figure 3.1: System Architecture

3.2. Feasibility Study

3.2.1. Economic Feasibility

Agentic Job Market Intelligence is designed to be deployable at near-zero infrastructure cost during the academic phase, with a clear path to production scalability.

Development Cost: All core technologies — FastAPI, React, PostgreSQL, Neo4j Community Edition, and Python — are open-source and free to use. The Intelligence Engine's algorithmic agents (Jaccard similarity, BFS, topological sort) operate without LLM API calls, keeping inference costs at zero for all core functionality. LLM integration is fully optional; the system degrades gracefully to template-based responses when no LLM provider is configured, eliminating mandatory API fees.

Deployment Cost: A Docker Compose configuration enables a complete local deployment with four containers (PostgreSQL, Neo4j, FastAPI, and Vite frontend) using a single command. Cloud deployment requires only a small VM (2 vCPU, 4 GB RAM) for development use. PostgreSQL and Neo4j Community Edition are both free; Neo4j Enterprise is only necessary at enterprise scale.

Data Cost: The system uses a custom-authored Kaggle dataset of 10,000 realistic job postings and the freely available O*NET skill taxonomy. A curated synonym map (synonyms.json) eliminates ongoing data-cleaning labor by automating skill name canonicalization.

Return on Investment:

- I. Reduction in career counseling overhead for academic institutions
- II. Personalized, data-driven guidance reduces misaligned educational investment
- III. Transparent, explainable recommendations improve student trust and engagement
- IV. Reusable multi-agent architecture can be extended to new datasets or institutions at marginal cost

3.2.2. Technical Feasibility

All technologies chosen for CareerGraph are production-proven, well-documented, and available as stable releases:

Table 3.1 : Technical Feasibility

Component	Technology	Version	Maturity
Web Framework	FastAPI	0.115.6	Production-grade ASGI
Graph Database	Neo4j	5	Industry standard, 10+ years
Relational Database	PostgreSQL	15	Industry standard, 30+ years
ORM & Migrations	SQLAlchemy + Alembic	2.0.36 / 1.14	Production-grade
Frontend	React + TypeScript	18.3.1 / 5.6.3	Industry standard
Containerization	Docker Compose	v2	Industry standard
LLM Integration	Anthropic SDK	0.40.0	Production-grade
Fuzzy Matching	rapidfuzz	3.11.0	C-extension, production-grade
Testing	pytest + pytest-asyncio	8.3.4 / 0.24.0	Industry standard

3.2.3. Operational Feasibility

For Students: The platform is web-based and accessible from any modern browser with no installation required. Profile creation takes under five minutes (skills, major, target roles). Recommendations and skill gap analyses are generated in real time (under two seconds). All recommendations include human-readable explanations, not just raw scores, improving student comprehension and trust.

For Administrators: A single protected endpoint (POST /admin/ingest/csv) handles all data ingestion. CSV-based ingestion accommodates the standard Kaggle dataset format. Fuzzy skill normalization reduces the burden of maintaining exact skill name mappings, with the NormalizationAgent handling synonym resolution automatically.

For Institutions: Docker Compose deployment allows IT staff to start the entire system with a single command. Environment-variable-based configuration (no hardcoded secrets) matches standard DevOps practices. Alembic database migrations provide reproducible, version-controlled schema setup across environments.

3.3. Requirement Analysis

3.3.1. Functional Requirements:

The functional requirements define what the system must do from the perspective of its users. The twelve requirements below were derived from the core use cases of the platform — student career guidance and administrator data management — and are prioritised by their impact on the system's primary value proposition. High-priority requirements cover the full intelligence pipeline (authentication, profile management, gap analysis, recommendations, roadmap, and market trends), while medium and low priorities address supporting features such as the dashboard, job explorer, and LLM-enriched explanations.

Table 3.2: Functional Requirements

ID	Requirement	Priority
FR-01	Users shall be able to register and log in using email and password	High
FR-02	Authenticated users shall be able to create and update a student profile with skills, major, graduation year, and target roles	High
FR-03	The system shall compute a weighted readiness score for a student against any target job based on their skill profile	High
FR-04	The system shall generate a ranked list of job recommendations using Jaccard similarity and partial skill overlap scoring	High
FR-05	The system shall identify skill gaps (missing and partially met skills) for any target job	High
FR-06	The system shall generate an ordered learning roadmap from the student's current skills to target job requirements via BFS and topological sort	High
FR-07	The system shall provide market demand intelligence showing in-demand skills and categories across all ingested jobs	High
FR-08	The system shall provide a dashboard with summary KPIs (readiness score, matched jobs, skill counts, missing high-demand skills)	Medium
FR-09	Administrators shall be able to ingest job data from CSV files via a protected endpoint	Medium
FR-10	The system shall provide LLM-generated natural language explanations for all algorithmic outputs when an LLM provider is configured	Medium

FR-11	The system shall provide a searchable, filterable job explorer with pagination	Medium
FR-12	The system shall expose a paginated skill catalog and market skill demand listing	Low

3.3.2. Non-Functional Requirements

The non-functional requirements define the quality attributes the system must satisfy regardless of specific features. They span six dimensions critical for an academic-grade platform: performance ensures recommendations are returned within two seconds; reliability guarantees the system remains fully operational even without an LLM provider; security protects user data through JWT gating, bcrypt password hashing, and parameterised queries; maintainability is enforced through a comprehensive test suite; and portability and scalability ensure the system can be deployed and extended with minimal friction.

Table 3.3 : Non - Functional Requirements

ID	Requirement	Measure
NFR-01	Performance — API response time for recommendation and gap analysis queries	< 2 seconds for up to 1,000 jobs
NFR-02	Reliability — System shall degrade gracefully when no LLM provider is configured	Template fallback always returns a valid, structured response
NFR-03	Security — All non-public endpoints shall require JWT authentication	11 of 13 endpoints require a valid Bearer token
NFR-04	Security — Passwords shall be stored as bcrypt hashes	bcrypt cost factor ≥ 12
NFR-05	Security — All database queries shall use parameterized inputs to prevent injection	Zero string-interpolated SQL or Cypher queries
NFR-06	Maintainability — Test coverage shall span all agents, routers, and services	87 tests across 19 test files, all passing
NFR-07	Portability — The full system shall run in Docker without host environment dependencies	Single docker-compose up startup

NFR-08	Scalability — Database connections shall use async pooling	asyncpg for PostgreSQL; neo4j native driver for Neo4j
NFR-09	Explainability — All recommendations shall include a human-readable explanation	Responses include matched skills, missing skills, and readiness score
NFR-10	Usability — UI shall be responsive and accessible from standard desktop browsers	React 18 + TailwindCSS + shadcn/ui

3.3.3. Tools & Technology

Backend

Table 3.4 : Tools (Backend)

Tool	Version	Purpose
Python	3.11	Core language
FastAPI	0.115.6	REST API framework (ASGI)
Uvicorn	0.32.1	ASGI production/dev server
Pydantic	2.10.4	Request/response validation and LLM output schemas
SQLAlchemy	2.0.36	ORM for PostgreSQL
Alembic	1.14.0	Database schema migrations
asyncpg	0.30.0	Async PostgreSQL driver
neo4j	5.26.0	Neo4j graph database driver
python-jose	3.3.0	JWT encoding and decoding
bcrypt	4.2.1	Password hashing
rapidfuzz	3.11.0	Fuzzy skill name matching (C-extension)
anthropic	0.40.0	Claude LLM SDK
openai	1.58.1	GPT-4o LLM SDK

pandas	2.2.3	CSV data processing for ingestion
pytest	8.3.4	Test framework
pytest-asyncio	0.24.0	Async test support

Frontend

Table 3.5 : Tools (Frontend)

Tool	Version	Purpose
React	18.3.1	UI component framework
TypeScript	5.6.3	Static type safety
Vite	6.0.5	Build tool and dev server
React Router	v6	Client-side routing
TanStack Query	5.62.7	Server-state caching
Axios	1.7.9	HTTP client
TailwindCSS	3.4.17	Utility-first CSS framework
shadcn/ui	Latest	Accessible component library

Infrastructure & Databases

Table 3.6 : Tools (Infrastructure & Databases)

Tool	Version	Purpose
PostgreSQL	15	Relational database — users, auth, profiles
Neo4j	5 (Community)	Graph database — knowledge graph
Docker	Latest	Containerization

Docker Compose	v2	Multi-service orchestration
python-dotenv	Latest	Environment variable management

Data Sources

Table 3.7 : Tools (Data Sources)

Source	Content	Use
Kaggle Job Dataset (kaggle_jobs.csv)	10,000 job postings: title, company, location, type, salary, skills	Primary job data for ingestion
O*NET Skill Taxonomy (onet_skills.csv)	Standard skill vocabulary and categories	Canonical skill names for normalization
Synonym Map (synonyms.json)	~500 skill name variants (e.g., ReactJS → React)	Bootstrap for fuzzy matching

3.4. System Design

3.4.1. Development Model

Agentic Job Market Intelligence follows an **Agile TDD (Test-Driven Development)** model executed in four sequential phases. Each phase begins with writing tests before any implementation — the canonical Red → Green → Refactor cycle.

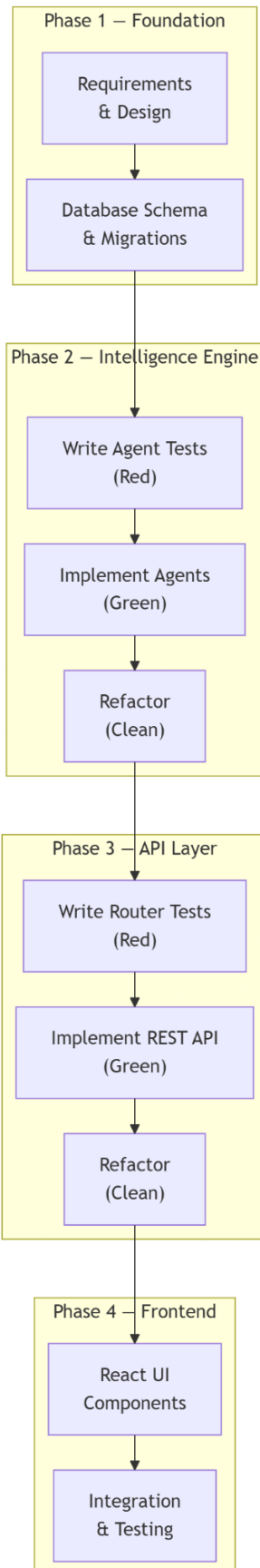


Figure 3.2 : Agile TDD Model

TDD Cycle:

1. Write a failing test that describes the required behavior
2. Implement the minimum code necessary to pass the test
3. Refactor for clarity without breaking tests
4. Repeat for the next requirement

This approach produced 87 unit and integration tests (all passing) before the frontend work began, ensuring a stable, reliable API contract for frontend integration.

3.4.2. Data Flow Diagram

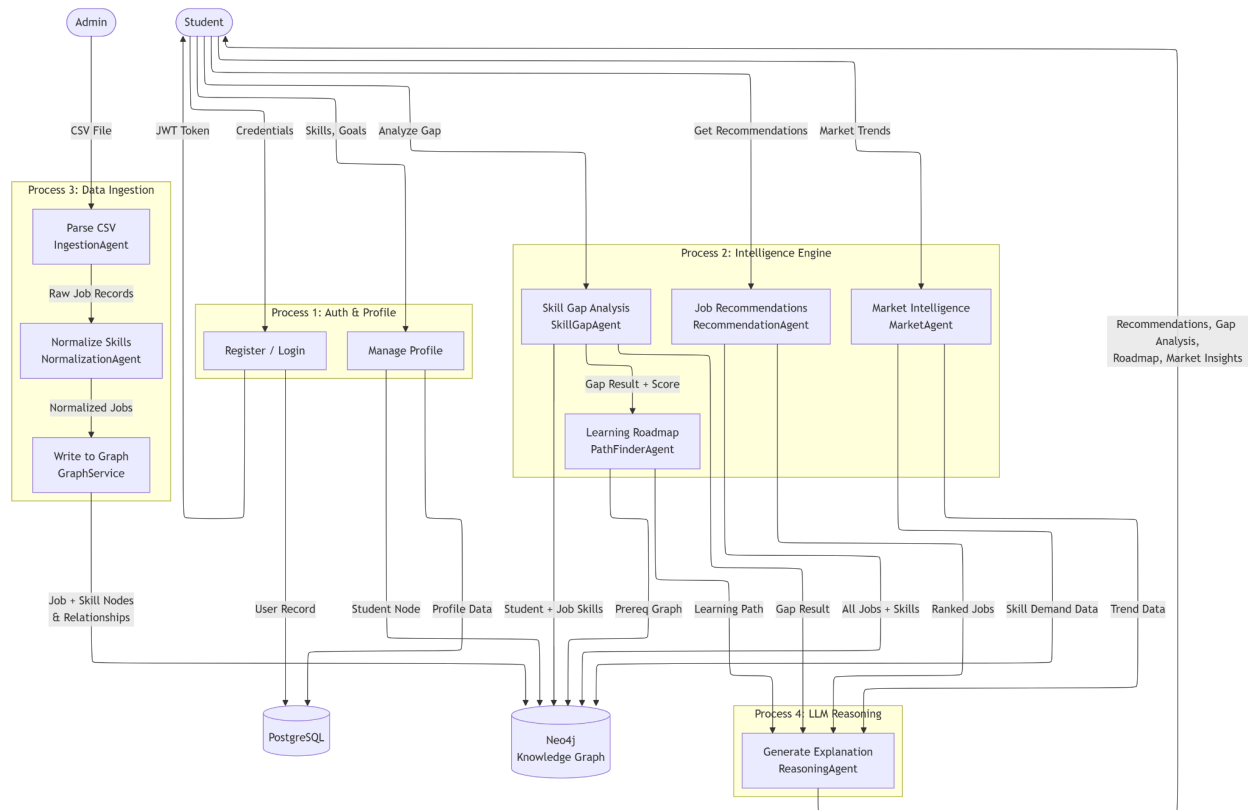


Figure 3.3 : Level 1 — Core Processes

The Level 1 diagram decomposes the system into four core processes: authentication and profile management, the intelligence engine, data ingestion, and LLM reasoning. Data flows illustrate

how student inputs trigger graph queries, and how algorithmic results are optionally enriched by the ReasoningAgent before being returned to the user.

The Level 2 diagram details the internal data flows of the skill gap analysis feature. The SkillGapAgent fetches the student's skills and the job's requirements from Neo4j and computes a weighted readiness score, which is then passed to the PathFinderAgent to generate an ordered learning path. The final response is either enriched by the ReasoningAgent or constructed from a structured template fallback.

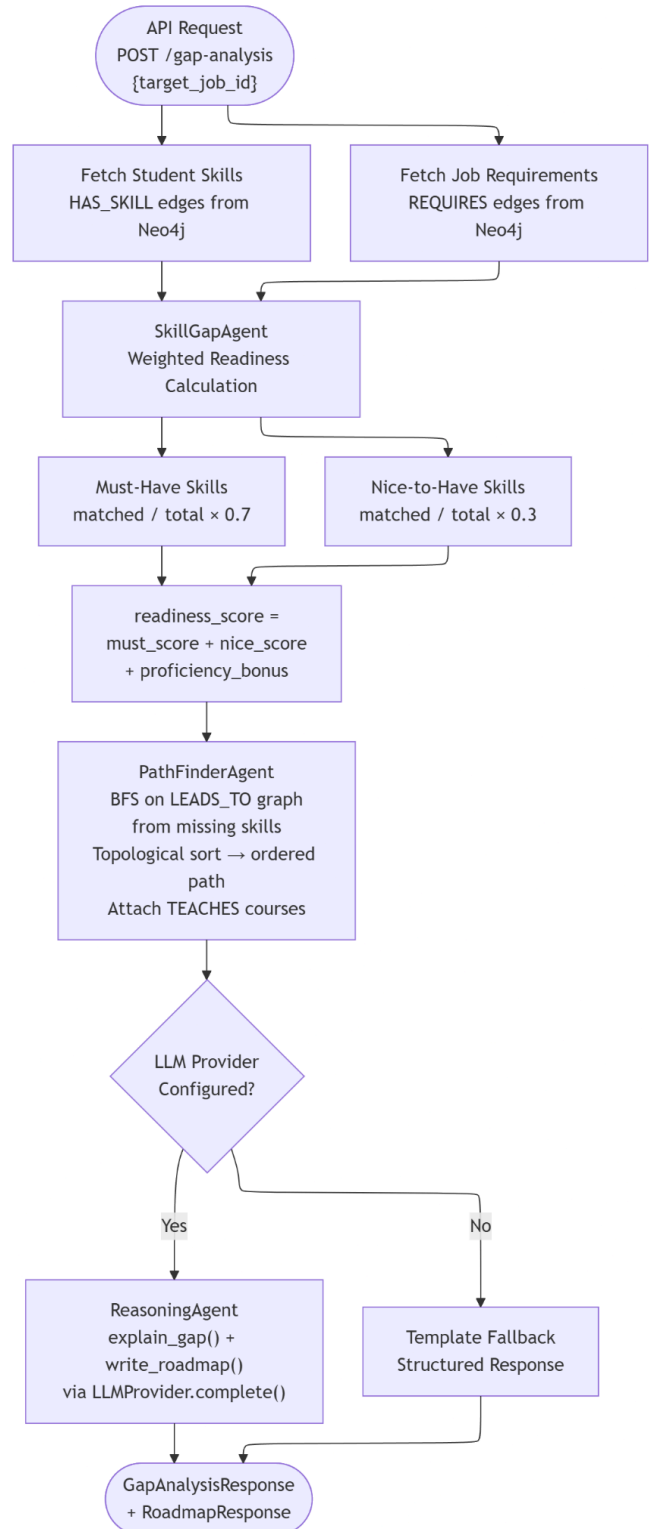


Figure 3.4 : Level 2 — Skill Gap Analysis Detail

3.4.3. Entity-Relationship Diagram

Note on dual-database design: USERS and STUDENT_PROFILES are stored in PostgreSQL for ACID-compliant relational operations. All other entities (SKILL, JOB, COURSE, CATEGORY) and their relationships exist in the Neo4j knowledge graph, where graph traversal queries enable prerequisite reasoning and skill proximity scoring.

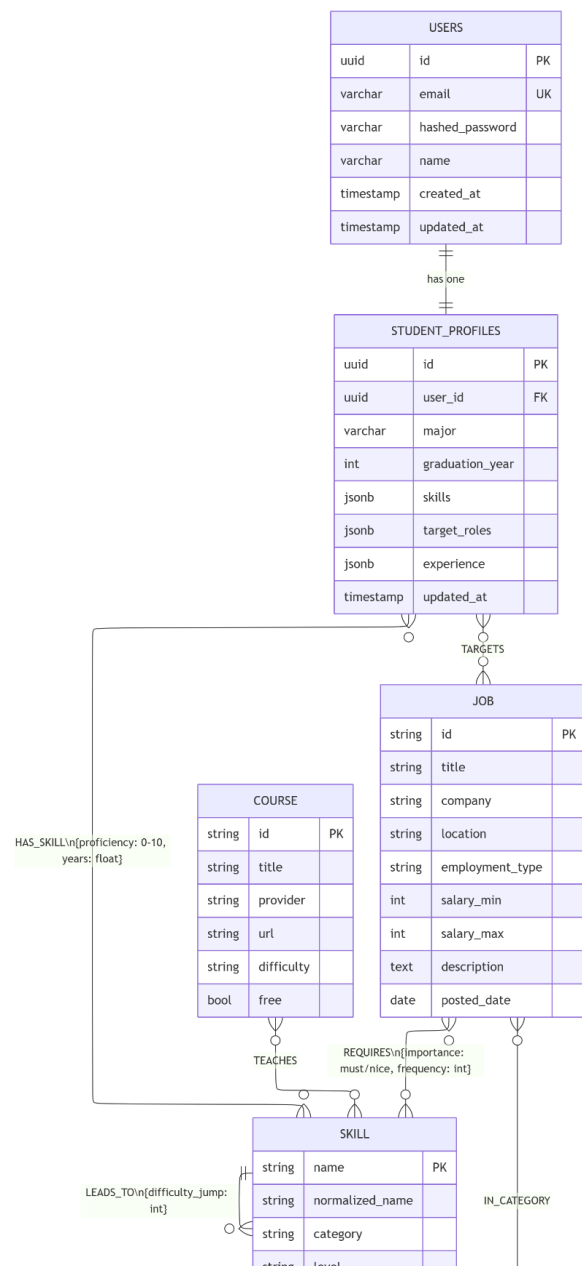


Figure 3.5 : Entity-Relationship Diagram

3.4.4. Database Schema Diagram

PostgreSQL — Relational Schema

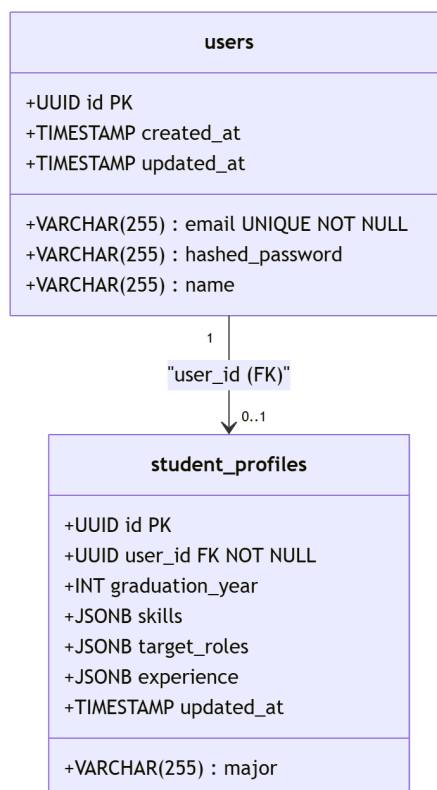


Figure 3.6 : PostgreSQL — Relational Schema

The relational schema stores only the data that requires ACID compliance: user credentials and student profiles. A one-to-one foreign key relationship links each authenticated user to their corresponding profile record in PostgreSQL.

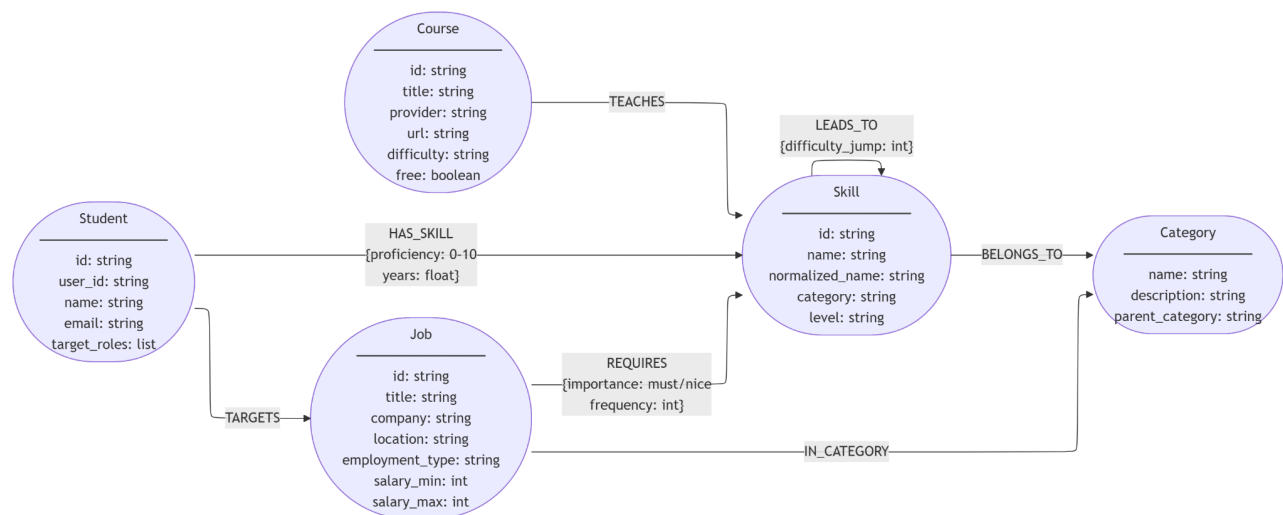


Figure 3.7 : Neo4j - Knowledge Graph Node & Relationship Schema

The relational schema stores only the data that requires ACID compliance: user credentials and student profiles. A one-to-one foreign key relationship links each authenticated user to their corresponding profile record in PostgreSQL.

Seed Data at Demo Scale / Full Scale

Table 3.8 : Seed Data

Node Type	Demo Seed	Full Dataset (Kaggle)
Student	3	— (user-created)
Job	50	10,000+
Skill	80	500+ (O*NET taxonomy)
Course	30	30
Category	7	20+

3.4.5. System Flowchart

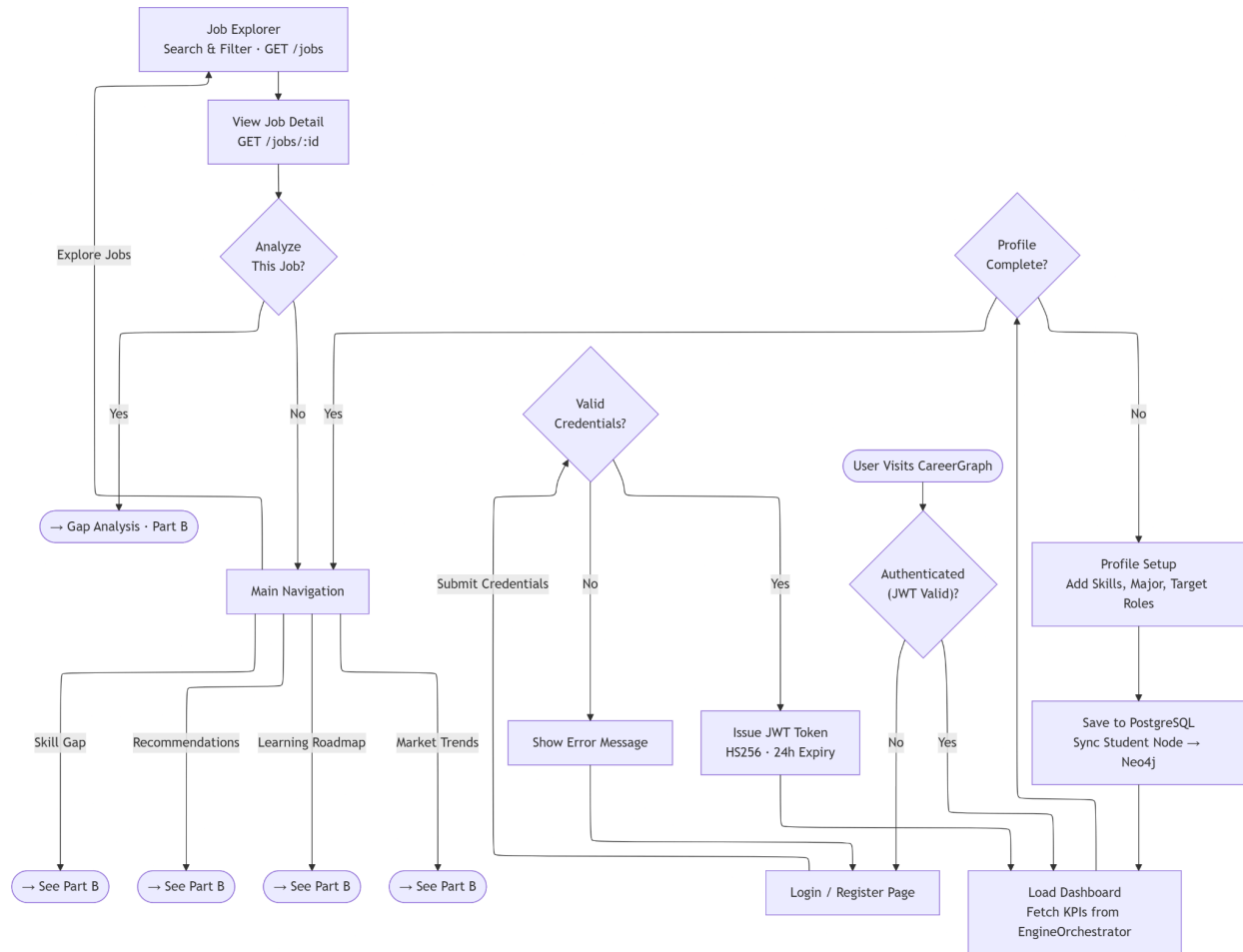


Figure 3.8 : System Flowchart — Part A: User Session & Job Exploration

Part A covers everything that happens before a student reaches the core intelligence features. When a user first visits the platform, the system checks whether a valid JWT token exists. If not, the user is directed to the login or registration page; on successful credential validation, a signed HS256 token with a 24-hour expiry is issued and the Dashboard is loaded. The Dashboard then checks whether the student's profile is complete — if skills, major, and target roles have not been filled in, the user is redirected to the Profile Setup page, which saves the data to PostgreSQL and simultaneously syncs a Student node into Neo4j. Once the profile is confirmed complete, the student reaches the Main Navigation hub, from which they can explore the Job Explorer or proceed to any of the four intelligence features covered in Part B. Within the Job Explorer, a student can browse and filter job listings and open a specific job's detail page; from there they may choose to trigger a gap analysis for that job, which hands off to the flow described in Part B.

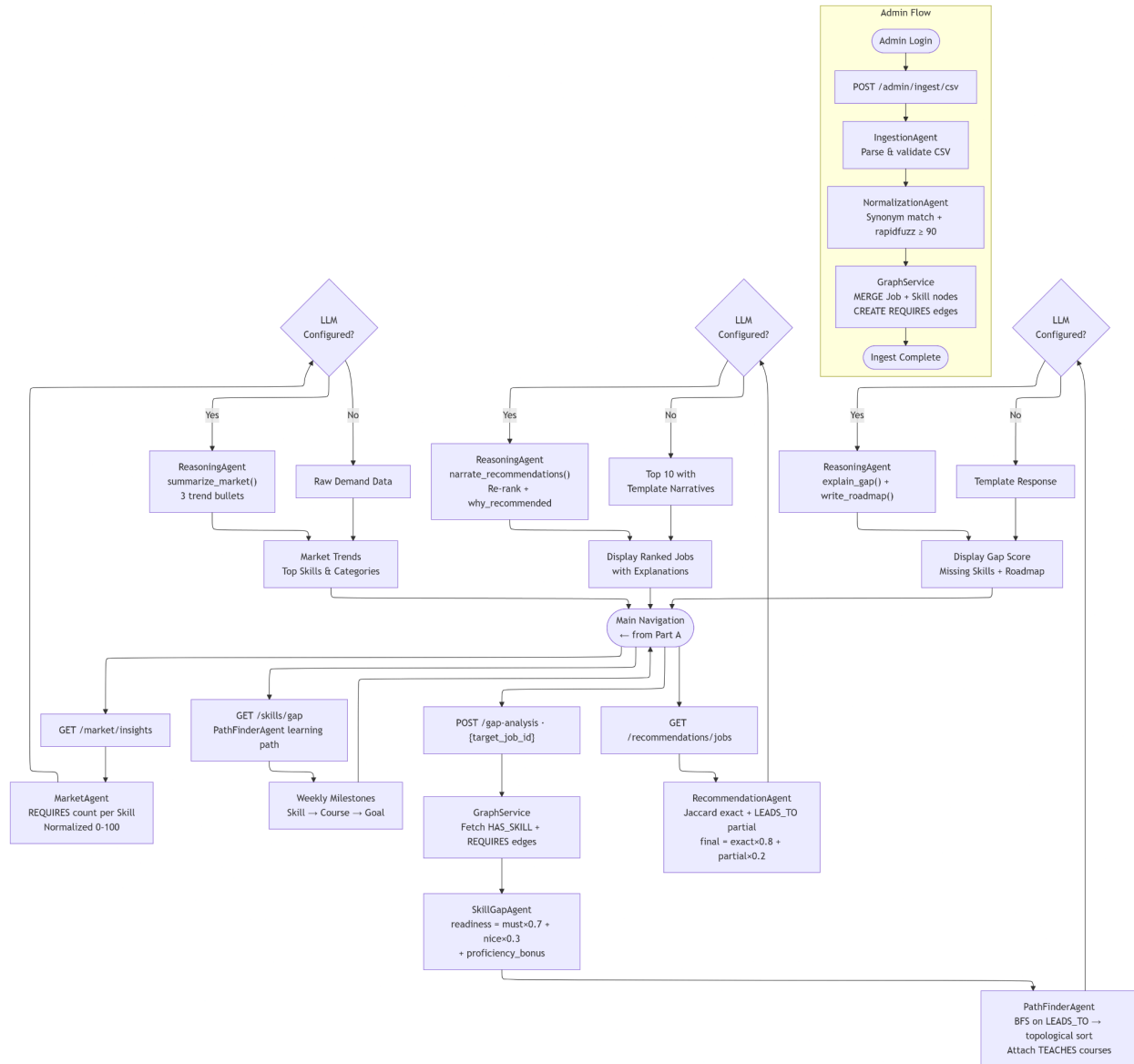


Figure 3.9 : System Flowchart — Part B: Intelligence Engine & Admin Flow

Part B covers the four intelligence features branching from the Main Navigation hub, plus the separate administrator ingestion flow. For recommendations, the RecommendationAgent scores every job using a composite Jaccard similarity (80% exact skill match, 20% LEADS_TO proximity) and returns the ranked results. For skill gap analysis, the SkillGapAgent computes a weighted readiness score against the target job, and the PathFinderAgent follows up with a BFS over the prerequisite graph to produce an ordered learning roadmap with attached courses. The market trends feature aggregates REQUIRES edge counts per skill across all jobs and normalises them to a 0–100 demand score. All four features share the same final step: results are either enriched with natural language by the ReasoningAgent when an LLM provider is configured, or returned as structured template responses. The admin flow runs independently — uploaded CSV

data is parsed by the IngestionAgent, skill names are normalised via synonym matching and fuzzy scoring, and the resulting job and skill nodes are written into the Neo4j knowledge graph.

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